

Lesson 4: Persistence and Streaming

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In [1]: from dotenv import load_dotenv
```

```
_ = load_dotenv()
```

```
In [2]: from langgraph.graph import StateGraph, END
from typing import TypedDict, Annotated
import operator
from langchain_core.messages import AnyMessage, SystemMessage, HumanMessage,
from langchain_openai import ChatOpenAI
from langchain_community.tools.tavily_search import TavilySearchResults
```

```
In [3]: tool = TavilySearchResults(max_results=2)
```

```
In [4]: class AgentState(TypedDict):
        messages: Annotated[list[AnyMessage], operator.add]
```

```
In [5]: from langgraph.checkpoint.sqlite import SqliteSaver

memory = SqliteSaver.from_conn_string(":memory:")
```

```
In [6]: class Agent:
        def __init__(self, model, tools, checkpointer, system=""):
            self.system = system
            graph = StateGraph(AgentState)
            graph.add_node("llm", self.call_openai)
            graph.add_node("action", self.take_action)
            graph.add_conditional_edges("llm", self.exists_action, {True: "action", False: "llm"})
            graph.add_edge("action", "llm")
            graph.set_entry_point("llm")
            self.graph = graph.compile(checkpointer=checkpointer)
            self.tools = {t.name: t for t in tools}
            self.model = model.bind_tools(tools)

        def call_openai(self, state: AgentState):
            messages = state['messages']
            if self.system:
                messages = [SystemMessage(content=self.system)] + messages
            message = self.model.invoke(messages)
            return {'messages': [message]}

        def exists_action(self, state: AgentState):
            result = state['messages'][-1]
            return len(result.tool_calls) > 0

        def take_action(self, state: AgentState):
            tool_calls = state['messages'][-1].tool_calls
            results = []
            for t in tool_calls:
                print(f"Calling: {t}")
```

```
        result = self.tools[t['name']].invoke(t['args'])
        results.append(ToolMessage(tool_call_id=t['id'], name=t['name'],
        print("Back to the model!")
        return {'messages': results}
```

```
In [7]: prompt = """You are a smart research assistant. Use the search engine to look up information. \
You are allowed to make multiple calls (either together or in sequence). \
Only look up information when you are sure of what you want. \
If you need to look up some information before asking a follow up question, \
"""
model = ChatOpenAI(model="gpt-4o")
abot = Agent(model, [tool], system=prompt, checkpoint=memory)
```

```
In [8]: messages = [HumanMessage(content="What is the weather in sf?")]
```

```
In [9]: thread = {"configurable": {"thread_id": "1"}}
```

```
In [10]: for event in abot.graph.stream({"messages": messages}, thread):
        for v in event.values():
            print(v['messages'])
```

```
[AIMessage(content='', additional_kwargs={'tool_calls': [{'id': 'call_GzBlwKx
Au2C0wFTR6n6tT4C', 'function': {'arguments': '{"query": "current weather in
San Francisco"}', 'name': 'tavily_search_results_json', 'type': 'function'
}]}], response_metadata={'token_usage': {'completion_tokens': 22, 'prompt_t
okens': 151, 'total_tokens': 173}, 'model_name': 'gpt-4o', 'system_fingerpri
nt': 'fp_319be4768e', 'finish_reason': 'tool_calls', 'logprobs': None}, id
='run-04c0500c-2f5c-4894-88f5-692ca92b5e29-0', tool_calls=[{'name': 'tavily_
search_results_json', 'args': {'query': 'current weather in San Francisco'},
'id': 'call_GzBlwKxAu2C0wFTR6n6tT4C'}]))]
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'current w
eather in San Francisco'}, 'id': 'call_GzBlwKxAu2C0wFTR6n6tT4C'}
Back to the model!
[ToolMessage(content='[{"url": "https://www.weatherapi.com/", "content
": "{\\"location\\": {\\"name\\": \\"San Francisco\\", \\"region\\": \\"California
\\", \\"country\\": \\"United States of America\\", \\"lat\\": 37.78, \\"lon\\": -12
2.42, \\"tz_id\\": \\"America/Los_Angeles\\", \\"localtime_epoch\\": 1718108084,
\\"localtime\\": \\"2024-06-11 5:14\\", \\"current\\": {\\"last_updated_epoch\\": 1
718107200, \\"last_updated\\": \\"2024-06-11 05:00\\", \\"temp_c\\": 11.7, \\"temp_
f\\": 53.1, \\"is_day\\": 0, \\"condition\\": {\\"text\\": \\"Partly cloudy\\", \\"ico
n\\": \\"//cdn.weatherapi.com/weather/64x64/night/116.png\\", \\"code\\": 1003},
\\"wind_mph\\": 6.9, \\"wind_kph\\": 11.2, \\"wind_degree\\": 300, \\"wind_dir\\":
\\"WNW\\", \\"pressure_mb\\": 1012.0, \\"pressure_in\\": 29.89, \\"precip_mm\\": 0.
0, \\"precip_in\\": 0.0, \\"humidity\\": 89, \\"cloud\\": 25, \\"feelslike_c\\": 10.
8, \\"feelslike_f\\": 51.5, \\"windchill_c\\": 10.5, \\"windchill_f\\": 50.9, \\"he
atindex_c\\": 11.3, \\"heatindex_f\\": 52.4, \\"dewpoint_c\\": 9.4, \\"dewpoint_f
\\": 49.0, \\"vis_km\\": 16.0, \\"vis_miles\\": 9.0, \\"uv\\": 1.0, \\"gust_mph\\":
9.3, \\"gust_kph\\": 14.9}}"}], {"url": "https://www.wunderground.com/weathe
r/us/ca/san-francisco", "content": "San Francisco Weather Forecasts. Wea
ther Underground provides local & long-range weather forecasts, weather repor
ts, maps & tropical weather conditions for the San Francisco area. ... 2024
(GMT -7 ...\\}'}], name='tavily_search_results_json', tool_call_id='call_GzBl
wKxAu2C0wFTR6n6tT4C')]
```

```
[AIMessage(content='The current weather in San Francisco is partly cloudy wi
th a temperature of approximately 11.7°C (53.1°F). The wind is blowing from
the west-northwest at about 6.9 mph (11.2 kph). The humidity is at 89%, and
the visibility is around 16 kilometers (9 miles). The UV index is low at
1.', response_metadata={'token_usage': {'completion_tokens': 78, 'prompt_tok
ens': 656, 'total_tokens': 734}, 'model_name': 'gpt-4o', 'system_fingerprin
t': 'fp_319be4768e', 'finish_reason': 'stop', 'logprobs': None}, id='run-a75
54d8d-1276-4540-a25f-a868d2c54fa7-0')]
```

```
In [11]: messages = [HumanMessage(content="What about in la?")]
thread = {"configurable": {"thread_id": "1"}}
for event in abot.graph.stream({"messages": messages}, thread):
    for v in event.values():
        print(v)
```

```
{'messages': [AIMessage(content='', additional_kwargs={'tool_calls': [{'id': 'call_kirMoamRpVXD480ZmUtbEkEE', 'function': {'arguments': '{"query": "current weather in Los Angeles"}', 'name': 'tavily_search_results_json', 'type': 'function'}]}], response_metadata={'token_usage': {'completion_tokens': 22, 'prompt_tokens': 746, 'total_tokens': 768}, 'model_name': 'gpt-4o', 'system_fingerprint': 'fp_319be4768e', 'finish_reason': 'tool_calls', 'logprobs': None}, id='run-17fe95b5-cbed-4143-8b94-733a27ad39ac-0', tool_calls=[{'name': 'tavily_search_results_json', 'args': {'query': 'current weather in Los Angeles'}, 'id': 'call_kirMoamRpVXD480ZmUtbEkEE'}])]}
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'current weather in Los Angeles'}, 'id': 'call_kirMoamRpVXD480ZmUtbEkEE'}
Back to the model!
{'messages': [ToolMessage(content='[{"url": "https://www.weatherapi.com/\n", "content": [{"location": {"name": "Los Angeles", "region": "California", "country": "United States of America", "lat": 34.05, "lon": -118.24, "tz_id": "America/Los_Angeles", "localtime_epoch": 1718108199, "localtime": "2024-06-11 5:16"}, "current": {"last_updated_epoch": 1718108100, "last_updated": "2024-06-11 05:15", "temp_c": 15.6, "temp_f": 60.1, "is_day": 0, "condition": {"text": "Mist", "icon": "//cdn.weatherapi.com/weather/64x64/night/143.png", "code": 1030}, "wind_mph": 2.2, "wind_kph": 3.6, "wind_degree": 10, "wind_dir": "N", "pressure_mb": 1012.0, "pressure_in": 29.89, "precip_mm": 0.0, "precip_in": 0.0, "humidity": 90, "cloud": 100, "feelslike_c": 15.6, "feelslike_f": 60.1, "windchill_c": 16.8, "windchill_f": 62.2, "heatindex_c": 16.8, "heatindex_f": 62.2, "dewpoint_c": 12.6, "dewpoint_f": 54.8, "vis_km": 8.0, "vis_miles": 4.0, "uv": 1.0, "gust_mph": 5.4, "gust_kph": 8.7}]}]', name='tavily_search_results_json', tool_call_id='call_kirMoamRpVXD480ZmUtbEkEE')]}
{'messages': [AIMessage(content='The current weather in Los Angeles is misty with a temperature of approximately 15.6°C (60.1°F). The wind is coming from the north at about 2.2 mph (3.6 kph). The humidity is at 90%, and the visibility is around 8 kilometers (4 miles). The UV index is low at 1.', response_metadata={'token_usage': {'completion_tokens': 75, 'prompt_tokens': 1287, 'total_tokens': 1362}, 'model_name': 'gpt-4o', 'system_fingerprint': 'fp_319be4768e', 'finish_reason': 'stop', 'logprobs': None}, id='run-5bf53fc2-f413-412f-a791-189f4b24538d-0')]}

```

```
In [12]: messages = [HumanMessage(content="Which one is warmer?")]
thread = {"configurable": {"thread_id": "1"}}
for event in abot.graph.stream({"messages": messages}, thread):
    for v in event.values():
        print(v)
```

```
{'messages': [AIMessage(content='Los Angeles is currently warmer than San Francisco. The temperature in Los Angeles is approximately 15.6°C (60.1°F), while in San Francisco it is around 11.7°C (53.1°F).', response_metadata={'token_usage': {'completion_tokens': 46, 'prompt_tokens': 1374, 'total_tokens': 1420}, 'model_name': 'gpt-4o', 'system_fingerprint': 'fp_319be4768e', 'finish_reason': 'stop', 'logprobs': None}, id='run-d42339b3-7194-4c57-bdeb-b3cb38ef000b-0')]}

```

```
In [13]: messages = [HumanMessage(content="Which one is warmer?")]
thread = {"configurable": {"thread_id": "2"}}
for event in abot.graph.stream({"messages": messages}, thread):
    for v in event.values():
        print(v)
```

```
{'messages': [AIMessage(content='I need more context to help answer your question. Are you comparing the temperature between two specific places, items, or times? Please provide more details.', response_metadata={'token_usage': {'completion_tokens': 31, 'prompt_tokens': 149, 'total_tokens': 180}, 'model_name': 'gpt-4o', 'system_fingerprint': 'fp_319be4768e', 'finish_reason': 'stop', 'logprobs': None}, id='run-b1fb941d-863e-43e2-a1da-6c794d756f23-0')]]}
```

Streaming tokens

```
In [14]: from langgraph.checkpoint.aiosqlite import AsyncSqliteSaver

memory = AsyncSqliteSaver.from_conn_string(":memory:")
abot = Agent(model, [tool], system=prompt, checkpointer=memory)
```

```
In [15]: messages = [HumanMessage(content="What is the weather in SF?")]
thread = {"configurable": {"thread_id": "4"}}
async for event in abot.graph.astream_events({"messages": messages}, thread,
    kind = event["event"]
    if kind == "on_chat_model_stream":
        content = event["data"]["chunk"].content
        if content:
            # Empty content in the context of OpenAI means
            # that the model is asking for a tool to be invoked.
            # So we only print non-empty content
            print(content, end="|")
```

```
/usr/local/lib/python3.11/site-packages/langchain_core/_api/beta_decorator.py:87: LangChainBetaWarning: This API is in beta and may change in the future.
```

```
warn_beta(
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'current weather in San Francisco'}, 'id': 'call_kveJTqztHP1Bs7Fm6FXGjPJA'}
Back to the model!
Currently|,| in| San| Francisco|,| the| weather| is| partly| cloudy| with| a|
| temperature| of| |11|.|7|°C| (|53|.|1|°F)|.| The| wind| is| blowing| from|
the| west|-n|orth|west| at| |6|.|9| mph| (|11|.|2| k|ph)|.| The| humidity| l
evel| is| at| |89|%.|
```

```
In [ ]:
```